

Exp 1 Given a magnetic wave in the far-field and in free-space

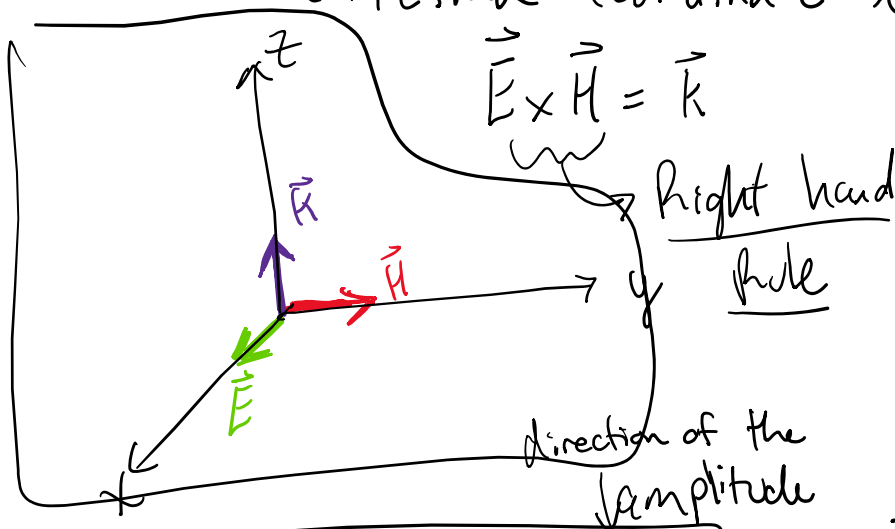
$$\vec{H} = \frac{E_0}{Z_0} \cos(\omega t \ominus kz) [\vec{u}_y] \left[ \frac{A}{m} \right]$$

Active Power  $\vec{E}$  and  $H$

$$\theta = 0^\circ$$

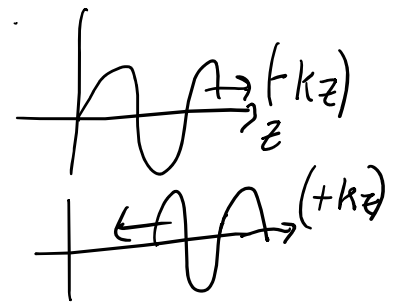
temporal

find the Electric wave eq. and plot its unit vectors on a cartesian coordinate system.



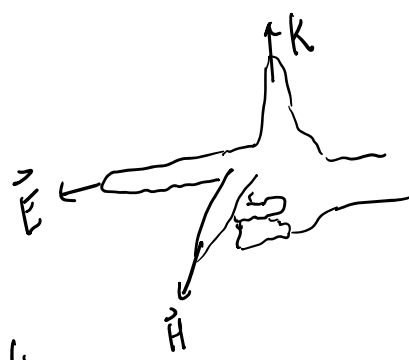
$$\vec{E} \times \vec{H} = \vec{k}$$

Right hand rule



$$\vec{E} = E_0 \cos(\omega t \ominus kz) [\vec{u}_x]$$

moving out from the (0,0,0)   
 propagation direction



Exp 2 E-field in free-space } we know the speed  
 $c = 3 \times 10^8 \text{ m/s}$

$$\vec{E} = 50 \cos\left(\underbrace{10^8 t}_{\omega = 2\pi f} + Kx\right) [\hat{a}_y] \left[\frac{V}{m}\right]$$

- 1.) find the direction of wave propagation  
• -x direction b/c  $(+Kx)$  in field equation.

- 2.) find  $K$  and the time it takes to travel  $\frac{\lambda}{2}$  distance.

$$K = \frac{2\pi f}{v} = \frac{10^8}{3 \times 10^8} = \frac{1}{3} \left[ \frac{\text{radians}}{m} \right]$$

$$K = \frac{1}{3} \frac{\text{rad}}{m}$$

$$f = \frac{c}{\lambda}$$

$$\rightarrow \frac{1}{t} = \frac{c}{\lambda}$$

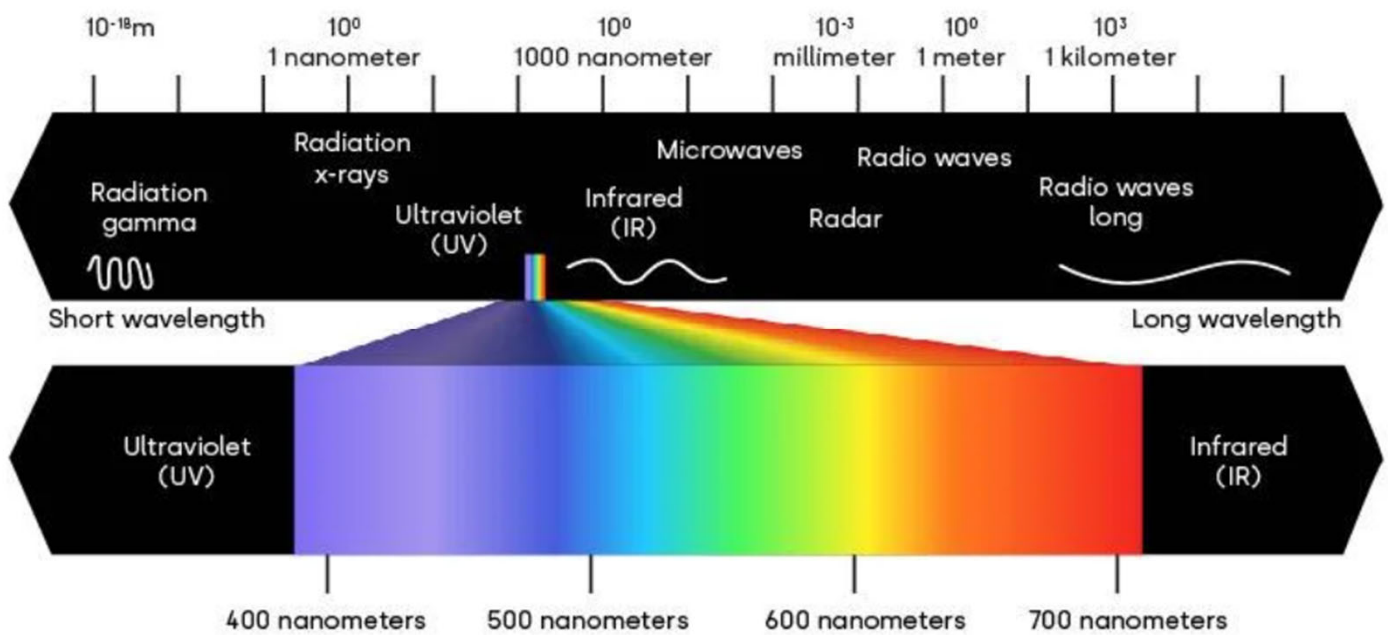
$$\rightarrow t = \frac{\lambda}{c}$$

$$t = \frac{1}{2} \frac{\lambda}{c}$$

$$\lambda = \frac{2\pi}{K}$$

$$t = \frac{1}{2} \left( \frac{2\pi}{Kc} \right)$$

$$t = \left( \frac{1}{2} \frac{6\pi}{c} \right) = 31.4 \text{ nsec}$$



Visible light Spectrum

